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Discrimination of phthalate species using a simple phage-based colorimetric sensor in conjunction with hierarchical support vector machine

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ABSTRACT

Here, the analytical potential of an M13 bacteriophage-based color sensor for discrimination of 4 phthalates with similar molecular structures (bis-(2-ethylhexyl)-phthalate (BEHP), dibutyl phthalate (DBP), diethyl phthalate (DEP), and benzyl-butyl-phthalate (BBP)) was investigated. The pattern and magnitude of the RGB color changes were different depending on the functional groups present in the phthalate structures. For example, BEHP possessing a long alkyl chain resulted in a minute color change, while the variation of color was substantially large when BBP containing an additional benzene ring was measured. Since a tryptophan-histidine-tryptophan residue possessing indole and imidazole was present on the self-assembled phages, the π-π interaction of benzene in BBP with the sensor surface produced a considerably greater color change. To evaluate the multi-modally varying color signals due to diverse interactions of the phthalates with the sensor and to discriminate them, support vector machine (SVM), which can construct a boundary hyperplane among complexly scattered sample groups, was used. In addition, hierarchical SVM (H-SVM) was adopted to deal with multi-class discrimination. The use of H-SVM improved the discrimination accuracy up to 90.1%, compared to 87.1% using SVM. The demonstrated color sensor is versatile and can be potentially adopted as an on-site screening tool. Strategies to improve the accuracy further for real applications are also discussed.

1. Introduction

Recently, the feasibility of an M13 bacteriophage-based color sensor whose structural color changes due to interactions of amino acid residues on the self-assembled phages with targeted gas molecules has been demonstrated for detection of various targets such as antibiotics (Moon et al., 2017a) and trinitrotoluene (TNT) (Oh et al., 2014) as well as cell recognition applications (Moon et al., 2017b). Contact with sample molecules caused the self-assembled M13 bacteriophage (length: ~880 nm, diameter: ~6.6 nm) bundle to swell and undergo a change of its structural color. Using the phage-based sensor is simple and involves just exposure to a gaseous sample for analysis. As a result, it is practical for use in diverse applications, especially simple on-site sensing.

As a potential environmental application of this color sensor, 4 different phthalates (bis-(2-ethylhexyl)-phthalate (BEHP), dibutyl phthalate (DBP), diethyl phthalate (DEP), and benzyl-butyl-phthalate (BBP)), which require attention as endocrine disrupting chemicals (EDCs) (Schettler, 2006; Waring and Harris, 2005; Witt and Sadik, 1999), were previously measured (Moon et al., 2016). This study showed somewhat different RGB color changes among these phthalates. However, the results were based on very simple and crude observations without demonstration of important analytical issues. For example, a proper discrimination method to treat the characteristic RGB signals and the accuracy of discrimination was not executed.

The aim of this study was to examine the analytical utility of a phage-based sensor for discrimination of BEHP, DBP, DEP, and BBP, and to provide their discrimination accuracies. The molecular structures of the phthalates are shown in Fig. 1. The samples have similar molecular structures possessing a common phthalate group with different side chain structures, complicating clear discrimination by a sensor. One important issue that requires consideration in this research is the use of a proper discrimination method to analyze the various RGB signals of the samples. Since the color changes from physical interaction

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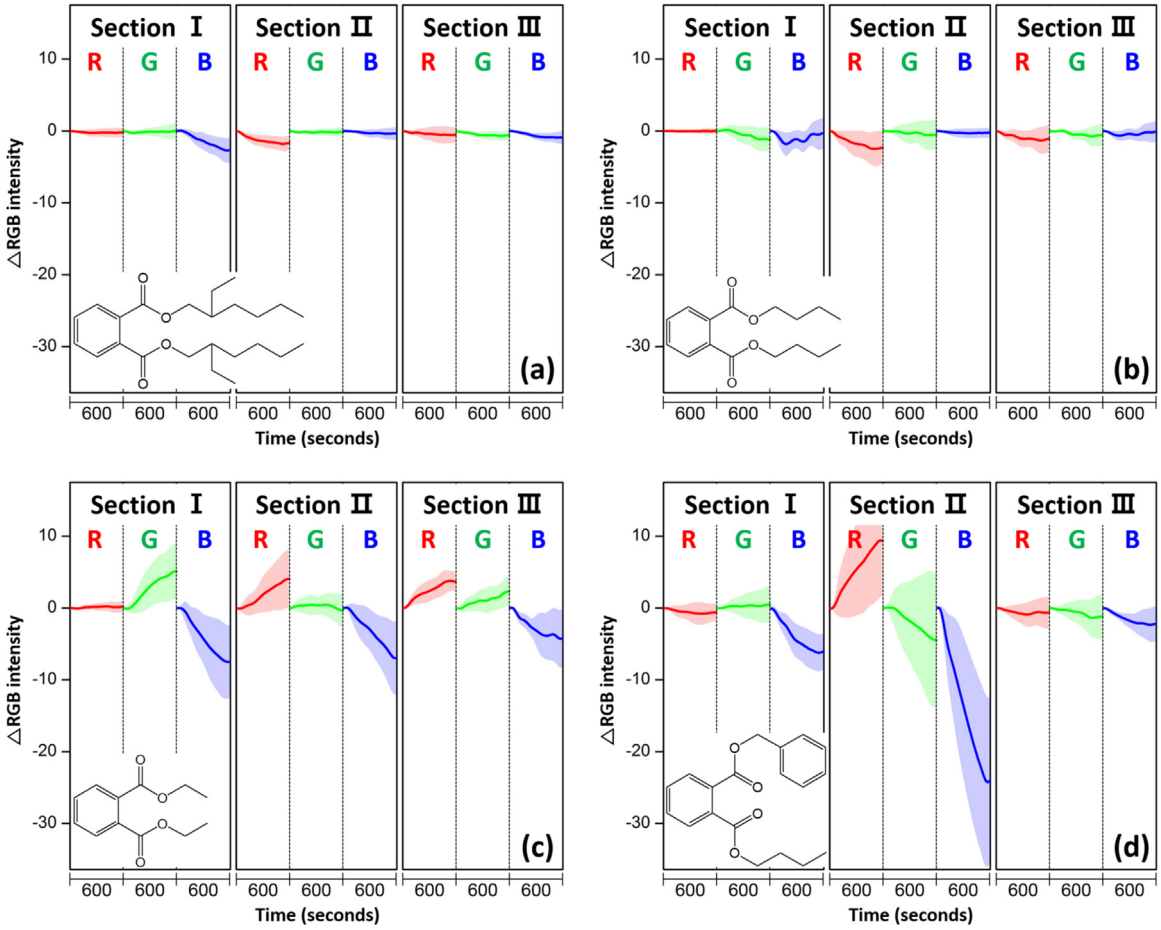

Fig. 1. Variations in average ΔRGB intensity obtained from the measurements of (a) BEHP, (b) DBP, (c) DEP, and (d) BBP. The shaded areas overlapping the corresponding average ΔRGB intensities indicate the standard deviations.

of an analyte with the self-assembled phages, the sensor is less specific to the molecular structures of samples compared to vibrational spectroscopic signatures. Therefore, a discrimination method able to effectively recognize relevant information in the RGB signal with low specificity will be beneficial. Also, the selected method needs to be able to classify multiple groups of the 4 different phthalates.

For data acquisition, vapors of the samples were measured using the sensors in a chamber, and the resulting variations of the RGB colors were recorded using a commonly-available CCD camera. The bacteriophages with repeating units of tryptophan-histidine-tryptophan were self-assembled on the surface (called the WHW color sensor) (Jaworski et al., 2011, 2008). Next, the magnitude and pattern of the varied RGB signals were analyzed in relation to the phthalate structures. It is reasonable to expect that color changes are not unimodal as the interactions between the sensor surface and the phthalates are multi-modal due to the different functional groups in each phthalate. Therefore, support vector machine (SVM) using a Lagrange multiplier (Burges, 1998; Cortes and Vapnik, 1995; Platt, 1998), a non-linear discrimination method, was employed for differentiation of phthalates based on the acquired RGB signals, and the resulting discrimination accuracies were evaluated. A more detailed description of SVM is provided in the experimental section.

Since 4 different phthalates need to be discriminated, a method effective in multi-group classification was required. For this purpose, hierarchical SVM (H-SVM) utilizing a hierarchical classifier in SVM (J.Weston, 1998; Koby Crammer, 2001; Schwenker, 2000) was further used for discrimination. The subsequent accuracy was compared to that obtained using just SVM. Finally, the dependency of discrimination error on the molecular structure of each phthalate was analyzed.

2. Material and methods

2.1. Fabrication of the M13 bacteriophage-based color sensor

WHW M13 bacteriophage-based color sensors were prepared by a simple pulling technique, as described earlier (Chung et al., 2011). A gold-coated Si wafer (525 μm total thickness, 100 nm-thick Au over a titanium adhesion layer; Platypus Technologies, Madison, WI, USA) was initially dipped in a WHW phage (concentration: 6 mg/mL) suspension in tri-buffered saline (12.5 mM Tris and 37.5 mM NaCl, pH 7.5). Then, by controlling the pulling speed between 10 and 100 mm/min using a programmable syringe pump (KD Scientific Co., LEGATO270), three sections (sections I, II, and III) with different self-assembly densities of phages were formed on each sensor. As a result, these three sections in each sensor showed dissimilar structural colors.

2.2. Preparation of samples and sample measurement by the color sensor

BEHP, DBP, DEP, and BBP were purchased from Sigma-Aldrich and used without further purification. Initially, a 1.0 L sealed glass chamber containing the color sensor on the top with a CCD camera (Celestron LCC., Torrance, CA, USA) and a heating block at the bottom to control the sample temperature was prepared. A sketch of the experimental apparatus is shown in the Supplementary data (Fig. S1). For the measurements, each sample (20 μL) was placed on the heating block, and the RGB signals at the three sections on the sensor were continuously acquired over 600 s while maintaining the sample temperature at 30 °C. There were no significant RGB signal changes after 600 s, and the elevation of temperature up to 60 or 90 °C did not enhance the color

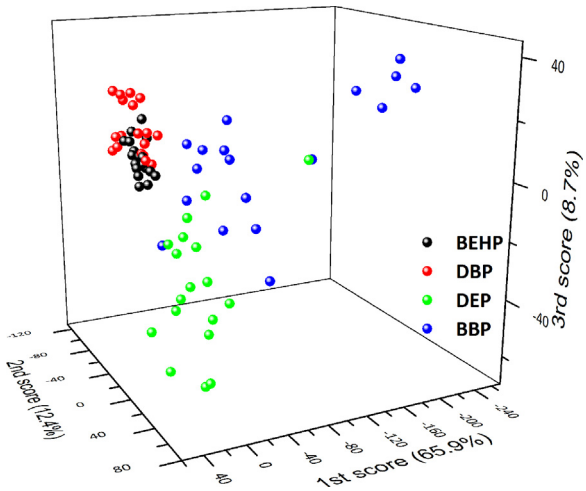

Fig. 2. Three-dimensional score scatter plot constructed using the first three scores obtained from all of the ΔRGB signals in the measurements of BEHP, DBP, DEP, and BBP.

change.

Three different color sensors were separately prepared, and 6 independent measurements were obtained using each sensor. Relative humidity in the chamber was maintained at approximately 20% during the measurements. A total of 18 RGB signals were acquired for each phthalate. The RGB differences (ΔRGB) between the control (no sample loaded) and sample measurements were calculated to compensate occasional fluctuation of humidity inside the chamber. ΔRGB signals were visualized using Photoshop (Adobe Systems Inc., San Jose, CA, USA).

2.3. SVM-based sample discrimination

SVM is a representative classification method among machine learning algorithms and has shown stable classification performance in diverse application fields (Team, 2005b). SVM maximizes the margins between two groups using a hyperplane and employs a Lagrange multiplier, which is a commonly used optimization technique, to estimate the hyperplane between the two groups. In practical classifications, target groups would be discriminable by either a linear plane or a nonlinear plane depending on the characteristics of the input data. Therefore, both linear and radial basis kernel functions were evaluated in this study. To determine the optimal parameters, the cost constant (C) was varied (1, 10, 100, and 1000) when using the linear kernel, and resulting mean square errors (MSEs) were evaluated. When the radial kernel was employed, both sigma (σ) and C were simultaneously varied (1, 10, 100, and 1000) to find the optimal combination yielding the lowest discrimination error. Both SVM and H-SVM were executed using the CRAN R program (library e1071: SVM package) (Dimitriadou et al., 2005; Team, 2005a). A detailed mathematical description of SVM is provided in the supplementary data.

3. Results and discussion

3.1. Examination of the RGB signals of the samples

The ΔRGB intensity corresponding to the magnitude of the color change relative to the color at the control measurement (no exposure of sample vapor) was evaluated. It is necessary to note again that tryptophan-histidine-tryptophan residue possessing indole and imidazole is present on the sensor surface. Also, the densities of the aligned phages in sections I, II, and III were different due to the various pulling speeds during self-assembly of the phages. Fig. 1(a) shows the variation in average ΔRGB intensities on sections I, II, and III over 600 s in the BEHP measurement. As described previously, 18 acquired ΔRGB intensities were averaged for each case. The shades overlapping the corresponding average ΔRGB intensities indicate the standard deviations of the observations. As shown in the figure, the color changes in each section upon exposure of BEHP are generally weak relative to the change of the blue color in section I. Therefore, it is reasonable to presume that the interaction between the sensor surface and BEHP is not strong, probably due to the greater hydrophobicity introduced by two ethylhexyl hydrocarbon chains.

Fig. 1(b), (c), and (d) show the variations in the average ΔRGB intensities of the measurements of DBP, DEP, and BBP, respectively. One very noticeable observation is that the magnitudes of the color variations become greater from DBP to DEP and then to BBP compared to BEHP. In the molecular structures of BEHP and DBP, the side chain of DBP is relatively less hydrophobic (2-ethylhexyl vs. dibutyl). Therefore, DBP is able to interact more attractively with the sensor surface, resulting in the greater color change. Accordingly, the color change in the measurement of DEP becomes considerably greater since the alkyl chain becomes shorter (from dibutyl to diethyl). In addition, the directions of the color changes become more apparent toward either the positive or negative direction. This demonstrates that the molecular interactions of DBP and DEP with the sensor surface are considerably dissimilar, although only their alkyl chain lengths are slightly different.

In the measurement of BBP, the color changes are also large in all three sections and exceptionally large in section II. The benzene ring instead of an alkyl chain in the BBP structure enhanced the π-π interaction with the benzene residue on sensor surface (the tryptophan-histidine-tryptophan residue), resulting in substantial color change. Overall, the magnitude of the color change was related to the structure of the side chain in the sample, and the increase of hydrophobicity produced a weaker color change. For reference, the final ΔRGB colors acquired at the final 600 s in each measurement are shown in the supplementary data (Fig. S2). Eighteen colored squares acquired from each measurement are provided.

To simply and simultaneously compare all of the RGB signals obtained from the samples, principal component analysis (PCA) was executed using the ΔRGB signals shown in Fig. 1, and the distribution of resulting scores was examined. Fig. 2 shows the three-dimensional (3D) score scatter plot constructed using the first three scores. The first, second, and third scores correspond to 65.9%, 12.4%, and 8.7% of the total variation present in the whole dataset, respectively. As shown in Fig. 2, the score distribution of the BBP samples is broadest due to the largest color change. Also, the scores of DEP are fairly spread out in the 3D domain. As expected, the scores of BEHP and DBP are narrowly distributed and overlap with each other. Overall, the score groups corresponding to the phthalates are somewhat discernible. Meanwhile,

Table 1
Obtained accuracies for discrimination of 4 different phthalates using SVM and H-SVM when either raw RGB signals or the scores are used as input data.

Input data	Classifier used	Accuracy (%)	Standard deviation (%)
Raw RGB signal	SVM	87.1	8.2
First 5 scores (93.4%)	SVM	80.1	9.6
First 10 scores (98.7%)	SVM	81.6	9.3
First 15 scores (99.6%)	SVM	81.3	9.1
First 20 scores (99.8%)	SVM	81.1	9.3
Raw RGB signal	H-SVM	90.1	7.1

Table 2
Accuracies resulting in discrimination of each phthalate in cases using SVM and H-SVM.

	BEHP	DBP	DEP	BBP
SVM	83.7%	76.7%	92.4%	94.6%
H-SVM	81.8%	90.4%	92.8%	95.6%

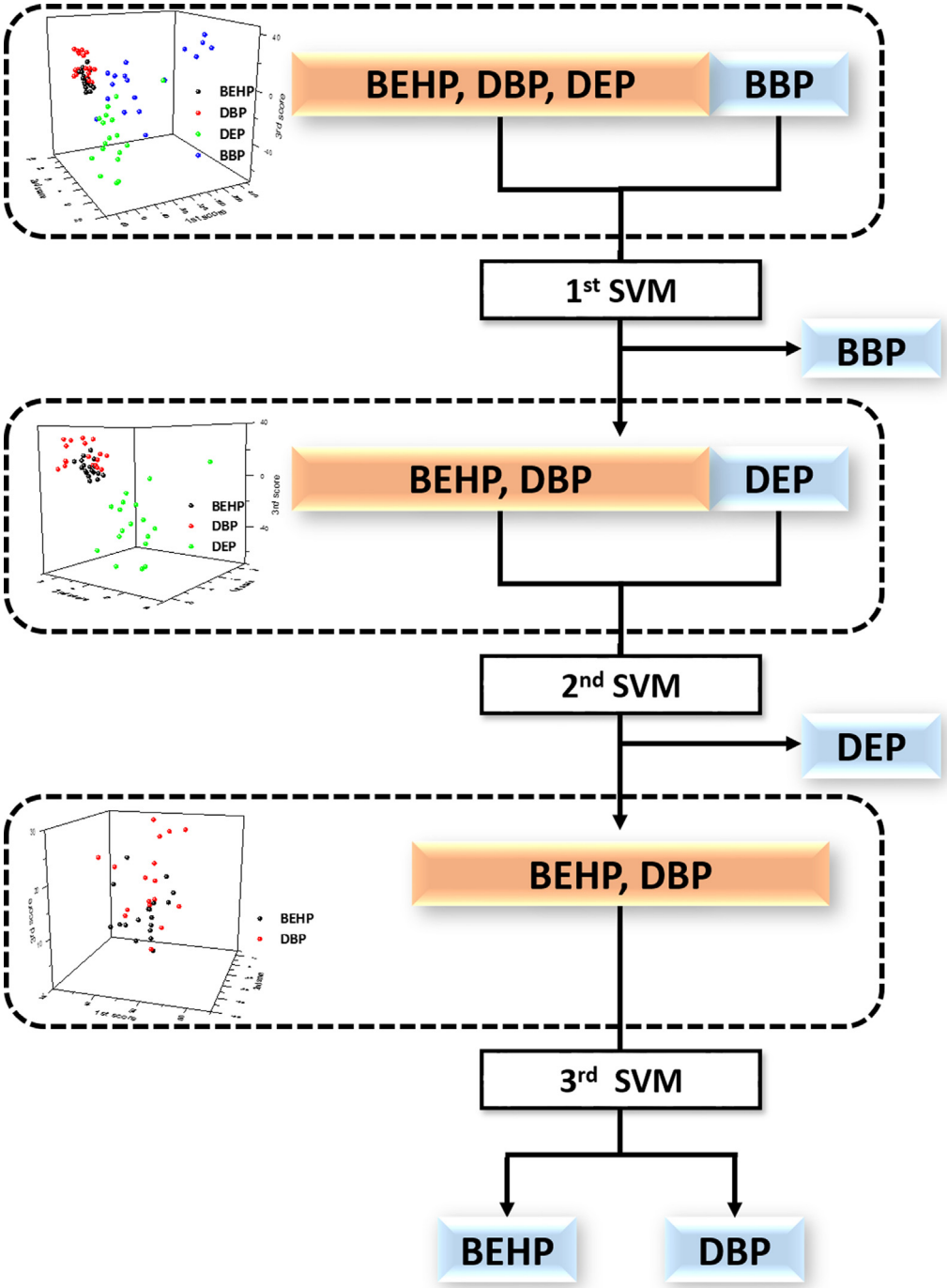

Fig. 3. Graphical description of the procedure of H-SVM for discrimination of 4 different phthalates (BEHP, DBP, DEP, and BBP). For better understanding, the targeted 3D score scatter plots at each step are also shown. The score scatter plots are provided for conceptual presentation only and were not actually used for discrimination.

some degree of overlap among the groups is inescapable, and the BEHP and DBP groups are more closely sited together. Therefore, a classifier able to effectively discriminate the groups is beneficial for accurate classification.

3.2. Evaluation of accuracy of discrimination of the 4 different phthalates

Since the molecular interaction with the tryptophan-histidine-tryptophan residue is expected to be multi-modal and complex due to the presence of several functional groups in each phthalate, the color does not vary linearly, as shown in Fig. 1. Therefore, SVM, a non-linear discrimination method, was employed for discrimination in this study. As input data for SVM, the ΔRGB signals without any treatment and the PCA scores obtained from them were utilized. For each case, the discrimination accuracy was obtained by cross-validating the whole dataset 1000 times with a random selection of 14 samples for training and using the remaining 4 samples for validation. The accuracy was obtained by calculating percentage of number of correctly classified samples over total number of samples. Table 1 shows the discrimination accuracies when either the raw color signals or the scores were used. The number in parentheses corresponds to the percentage of signal change explained by the employed scores from the total variation. The

use of raw color signal was better for discrimination compared to using PCA scores, and the resulting accuracy was 87.1% when the radial kernel (σ: 1000, *C*: 10) was used. This implies that the raw color signal contains more detail and descriptive information for discrimination. Meanwhile, the use of PCA scores is advantageous for representing the overall color signal with a much smaller number of variables (scores). However, the detailed features necessary to differentiate samples with similar molecular structures would not be fully highlighted in the course of PCA. Also, the use of more than 10 scores did not noticeably improve the accuracy.

The accuracies resulting in discrimination of each phthalate were further analyzed, as shown in Table 2. As reasonably expected, the discrimination of BBP is most accurate due to the distinct color change, and DEP is also easily differentiable. The discriminations of BEHP and DBP were relatively inaccurate and considerably below the overall accuracy (87.1%). This indicates that SVM is not sufficient to discriminate all of the phthalates with good accuracy. Therefore, an alternative approach able to effectively handle multi-class discrimination is required.

For this purpose, H-SVM to classify multiple groups in a stepwise fashion was used in this study. Since SVM is a binary classifier, for multi-class classification, one-vs-all classification approach was used. Fig. 3 graphically describes the H-SVM procedure. For better understanding, the targeted 3D score scattering plots at each step are also shown. The score scatter plots are provided for conceptual presentation only and were not actually used for discrimination. When classification of *n* groups (4 groups here) is encountered using H-SVM, a first SVM model (equivalent to a first hyperplane) is initially built to maximize the discrimination of the most distinguishable group from the other groups. Next, the group chosen first is excluded, and a second SVM model is built to discriminate the most distinguishable group out of the remaining groups (*n*-1 groups). This procedure proceeds *n*-1 times until all groups are individually discriminated. This corresponds to the use of *n*-1 individual SVM hyperplanes (models) for multi-class discrimination.

For execution of H-SVM, it is necessary to determine the classification hierarchy, namely which group will be the most distinguishable from the other groups. The order was determined by the SVM results. As shown in Table 2, BBP was the most discernable, followed by DEP. Therefore, as shown in Fig. 3, H-SVM was serially executed for discrimination of BBP vs. BEHP-DBP-DEP, DEP vs. BEHP-DBP, and BEHP vs. DBP (cross-validation 1000 times in each case). The use of H-SVM increased the accuracy up to 90.1%, and the improvement was statistically significant compared to that (87.1%) obtained using the SVM. In each SVM model, the radial kernel (σ: 1000, *C*: 10) was also used.

Next, the accuracies resulting in discrimination of each phthalate were examined (Table 2). Most notably, the identification of DBP became much more accurate compared to the result obtained using SVM (from 76.7% to 90.4%). The use of a separate hyperplane only for discrimination between DBP and BEHP explains the improved accuracy. In contrast, the discrimination of either BBP or DEP was minutely improved. Since these two phthalates exhibited distinct color changes, their discriminations were rather straightforward regardless of using either SVM or H-SVM. Overall, H-SVM was more effective for discrimination of all of the phthalates, and the achieved overall discrimination accuracy (90.1%) using H-SVM was greatly impacted by the improvement in discrimination between DBP and BEHP.

4. Conclusion

The analytical potential of a bacteriophage-based color sensor was demonstrated for discrimination of four different phthalates with similar molecular structures. H-SVM was adequate to handle the RGB signals, which multi-modally varied due to diverse interactions of the phthalates with the amino acids on the sensor, and was ultimately effective in discriminating multiple groups. Nonetheless, further efforts are necessary to improve the discrimination accuracy to allow the sensor to be utilized in real field applications. One attractive way to realize this goal is to use an array of multiple color sensors constructed with several phages possessing different amino acid residues, such as glutamic acid-glycine-aspartic acid and four repeating unit of glutamates. The dissimilar interactions of phthalates with each employed sensor yield various color changes, and more diverse information is useful to improve the discrimination. Finally, use of the proposed multi-sensor array for analysis of real samples and examination of related scientific issues is underway.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.envres.2018.12.030.

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